



UNSW
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MATH1231 Mathematics 1B – Algebra

Section 04 - Linear Combinations and Spans

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Linear combinations

Definition (Linear combination)

A **linear combination** of vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ in a vector space V is any expression of the form

$$\lambda_1 \vec{v}_1 + \lambda_2 \vec{v}_2 + \dots + \lambda_n \vec{v}_n,$$

where $\lambda_1, \lambda_2, \dots, \lambda_n$ are scalars.

Note that if $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ are in a vector space V , then every linear combination of these vectors is also in V . (Why?)

Examples 1.

a) We have $\begin{pmatrix} -1 \\ -3 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} - 3 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \end{pmatrix}$
so $\begin{pmatrix} -1 \\ -3 \end{pmatrix}$ is a linear combination of $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$.

b) Also, $7 - 2x + 2x^2 = 3(-1 + x + 4x^2) + 5(2 - x - 2x^2)$,
so $7 - 2x + 2x^2$ is a linear combination of the polynomials $-1 + x + 4x^2$ and $2 - x - 2x^2$.

Linear combinations, continued

Example 2. Is $\mathbf{0} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ a linear combination of $\mathbf{v}_1 = \begin{pmatrix} 9 \\ -1 \\ 6 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} \sqrt{2} \\ 3 \\ -\sqrt{3} \end{pmatrix}$

and $\mathbf{v}_3 = \begin{pmatrix} -\sqrt{3} \\ \sqrt{2} \\ 5 \end{pmatrix}$?



Linear combinations, continued

Example 3. Is $\mathbf{u} = \begin{pmatrix} 6 \\ 2 \\ 1 \end{pmatrix}$ a linear combination of $\mathbf{u}_1 = \begin{pmatrix} 2 \\ 0 \\ -5 \end{pmatrix}$, and $\mathbf{u}_2 = \begin{pmatrix} 0 \\ 1 \\ 7 \end{pmatrix}$?



Solution of example 3.

$\vec{u}$ is a linear combination of $\vec{u}_1$ and $\vec{u}_2$
 iff there exists $d, \mu \in \mathbb{R}$ such that

$$\vec{u} = d\vec{u}_1 + \mu\vec{u}_2 \quad \text{i.e.} \quad \begin{pmatrix} 6 \\ 2 \\ 1 \end{pmatrix} = d \begin{pmatrix} 2 \\ 0 \\ -5 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \\ 7 \end{pmatrix} \quad (*)$$

The augmented matrix is $M = \left(\begin{array}{cc|c} 2 & 0 & 6 \\ 0 & 1 & 2 \\ -5 & 7 & 1 \end{array} \right)$

$$R_1 \leftarrow \frac{1}{2} R_1 \rightarrow \left(\begin{array}{cc|c} \boxed{1} & 0 & 3 \\ 0 & 1 & 2 \\ -5 & 7 & 1 \end{array} \right) \xrightarrow[\text{R}_3 := \text{R}_3 + 5\text{R}_1]{\text{R}_3 := \text{R}_3 + 5\text{R}_1} \left(\begin{array}{cc|c} \boxed{1} & 0 & 3 \\ 0 & \boxed{1} & 2 \\ 0 & 7 & 16 \end{array} \right) \xrightarrow[\text{R}_3 := \text{R}_3 - 7\text{R}_2]{\text{R}_3 := \text{R}_3 - 7\text{R}_2}$$

$$R_3 := R_3 - 7R_2 \rightarrow \left(\begin{array}{cc|c} \boxed{1} & 0 & 3 \\ 0 & \boxed{1} & 2 \\ 0 & 0 & \boxed{2} \end{array} \right) \leftarrow \text{No solution}$$

$\therefore \vec{u}$ is not a linear combination of $\vec{u}_1$ and $\vec{u}_2$
Method 2 By (*) $d=3$; $\mu=2$ & this is inconsistent with the 3rd equation.

No: the equation

$$(6, 2, 1)^T = \lambda_1(2, 0, -5)^T + \lambda_2(0, 1, 7)^T$$

is easily seen to have no solution.

Linear combinations, continued

Example 4. Consider the matrices $A_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $A_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $A_3 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$.

- a) Is the matrix $B = \begin{pmatrix} -1 & 5 \\ 5 & 0 \end{pmatrix}$ a linear combination of the matrices A_1 , A_2 and A_3 ?
- b) Is the matrix $C = \begin{pmatrix} 2 & 9 \\ 3 & 1 \end{pmatrix}$ a linear combination of the matrices A_1 , A_2 and A_3 ?
- c) Describe all linear combinations of A_1 , A_2 , A_3 .



Example 4. Consider the matrices $A_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $A_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $A_3 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$.

a) Is the matrix $B = \begin{pmatrix} -1 & 5 \\ 5 & 0 \end{pmatrix}$ a linear combination of the matrices A_1 , A_2 and A_3 ?

▷ YES (It is a symmetric matrix)

b) Is the matrix $C = \begin{pmatrix} 2 & 9 \\ 3 & 1 \end{pmatrix}$ a linear combination of the matrices A_1 , A_2 and A_3 ?

▷ NO (not a symmetric matrix)

c) Describe all linear combinations of A_1 , A_2 , A_3 .

▷ It is the set of symmetric matrices.

Span

Definition (Span)

The **span** of a set of vectors $S = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ in a vector space V , written

$$\text{span}(S) \text{ or } \text{span}(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n),$$

is the set of all linear combinations of these vectors.

Note that $S \subseteq \text{span}(S) \subseteq V$.

Example 5.

a. Describe geometrically the span of $S = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}$.

Solution. Here $\text{span}(S) = \left\{ \lambda \begin{pmatrix} 1 \\ 2 \end{pmatrix} : \lambda \in \mathbb{R} \right\}$ is the line through the origin and the point $(1, 2)$ in $\mathbb{R}^2$.

Example 5: Examples of spans, continued.

b. Describe geometrically the span of $S = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right\}$.

Solution. Here $\text{span}(S) = \left\{ \lambda \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 3 \end{pmatrix} : \lambda, \mu \in \mathbb{R} \right\}$ is the plane through the origin and the points $(1, 2)$ and $(2, 3)$ in $\mathbb{R}^2$ (that is, all of $\mathbb{R}^2$).

c. Describe geometrically the span of $S = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \end{pmatrix} \right\}$.

Solution. Here $\text{span}(S) = \left\{ \lambda \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 4 \end{pmatrix} : \lambda, \mu \in \mathbb{R} \right\}$
$$= \left\{ (\lambda + 2\mu) \begin{pmatrix} 1 \\ 2 \end{pmatrix} : \lambda, \mu \in \mathbb{R} \right\}$$

is the line through the origin and the point $(1, 2)$ in $\mathbb{R}^2$.

d. Let $A_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $A_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $A_3 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. What is $\text{span}(A_1, A_2, A_3)$ in $M_{2,2}(\mathbb{R})$?

Solution. $\text{span}(A_1, A_2, A_3)$ is the set of all symmetric 2×2 matrices with real entries, because taking linear combinations of the given matrices will give *all* symmetric 2×2 matrices *and no others*.

Example 5: Examples of spans, continued.

- e. Consider $S = \{1, x, x^2\}$, a subset of $\mathbb{P}(\mathbb{R})$.

(Recall that $\mathbb{P}(\mathbb{R})$ is the set of all polynomials with real coefficients.)

- (i) What is $\text{span}(S)$?

Solution. Any linear combination of S is of the form $a_0 + a_1x + a_2x^2$ for some $a_0, a_1, a_2 \in \mathbb{R}$. Since $\mathbb{P}_2$ is the set of all polynomials of this form, we can say $\text{span}(S) = \mathbb{P}_2$.

Note: We used $\mathbb{P}_2$ as shorthand for $\mathbb{P}_2(\mathbb{R})$ here. We shall often write just $\mathbb{P}$ instead of $\mathbb{P}(\mathbb{F})$ if it is clear from the context what scalar field we are talking about.

- (ii) Is x^3 in the span of $S = \{1, x, x^2\}$?

Solution. Here $x^3 \notin \text{span}(S)$, since x^3 cannot be written as a linear combination $a_0 + a_1x + a_2x^2 = x^3$ for any $a_0, a_1, a_2 \in \mathbb{R}$.

Alternatively, we could say given the previous example that $x^3 \notin \mathbb{P}_2$.

The span of anything is a subspace

In the last four examples, the span of various numbers of vectors has turned out to be a vector space. In fact, this is always true.

Fact (Spans are subspaces)

Let V be a vector space; let $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ be vectors in V , and write $W = \text{span}(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n)$. Then W is a subspace of V . In fact, W is the *smallest* subspace of V which contains $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$.

Proof. We use the subspace theorem.

◇ As we mentioned earlier, every linear combination of vectors in V is also in V ; thus $\text{span}(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n)$ is a subset of V ; and it is given that V is a vector space.

◇ The zero vector $\vec{0}$ is in W because it is a linear combination of the given vectors, $\vec{0} = 0\vec{v}_1 + 0\vec{v}_2 + \dots + 0\vec{v}_n$.

◇ Now suppose that $\vec{w}$ and $\vec{x}$ are in W . By the definition of span, we have

$$\vec{w} = \lambda_1 \vec{v}_1 + \lambda_2 \vec{v}_2 + \dots + \lambda_n \vec{v}_n \quad \vec{x} = \mu_1 \vec{v}_1 + \mu_2 \vec{v}_2 + \dots + \mu_n \vec{v}_n$$

for some scalars $\lambda_1, \lambda_2, \dots, \lambda_n, \mu_1, \mu_2, \dots, \mu_n$. By using the associative, commutative and distributive laws in V we obtain

$$\vec{w} + \vec{x} = (\lambda_1 + \mu_1) \vec{v}_1 + (\lambda_2 + \mu_2) \vec{v}_2 + \dots + (\lambda_n + \mu_n) \vec{v}_n,$$

which is a linear combination of $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ and is therefore in W . Thus W is closed under addition.

The span of anything is a subspace, continued

◇ Finally, let $\vec{w}$ be in W and let λ be a scalar. Then

$$\vec{w} = \lambda_1 \vec{v}_1 + \lambda_2 \vec{v}_2 + \cdots + \lambda_n \vec{v}_n$$

for some scalars $\lambda_1, \lambda_2, \dots, \lambda_n$, and so

$$\lambda \vec{w} = (\lambda \lambda_1) \vec{v}_1 + (\lambda \lambda_2) \vec{v}_2 + \cdots + (\lambda \lambda_n) \vec{v}_n,$$

which again is a linear combination of $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$. Therefore W is closed under scalar multiplication.

We have shown that W is a subset of the given vector space V , and that W contains the zero vector and is closed under addition and scalar multiplication. Hence, W is a subspace of V .

◇ To show that W is the *smallest* subspace of V containing $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$, let X be *any* subspace of V containing these vectors: we want to show that W is a subset of X . But this is true because every linear combination of $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ is in X , and so every element of W is in X . This completes the proof.

Yet another way to prove something is a vector space

TIP!

Besides its theoretical importance, the previous result can be used to provide a very short and efficient proof that a given set is a vector space. We re-do a previous example.

Example 6. Show that $S = \{(x_1, 2x_2, x_2, x_1 + x_2) \mid x_1, x_2 \in \mathbb{R}\}$ is a vector space.

Solution. We have $S = \{x_1(1, 0, 0, 1) + x_2(0, 2, 1, 1) \mid x_1, x_2 \in \mathbb{R}\}$
 $= \text{span}\{(1, 0, 0, 1), (0, 2, 1, 1)\}$

and this is a vector space because the span of *any* (finite) collection of vectors is a vector space.

Spanning sets

Next, we want to approach the previous problem from a slightly different point of view. Given $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ in a vector space V , we have been trying to find out exactly what $\text{span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ is.

Problem

Now we only want to know if $\text{span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ equal to the whole vector space V , or just a part of V .

Definition (spanning set)

For vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ in a vector space V , the following statements all mean the same thing:

- every vector in V is a linear combination of $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$;
- $\text{span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\} = V$;
- $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ is a **spanning set** for V ;
- $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ **spans** V .

The point is: we know that $\text{span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ is a subspace of V ; this subspace may be the whole of V , or it may be only part of V . The former case is what is meant by saying “ $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ spans V ”.

Spanning sets for $\mathbb{R}^3$

Problem

Do *two* vectors span $\mathbb{R}^3$? Sometimes? Always? Never?

Never. Two vectors $\vec{v}, \vec{w}$ in $\mathbb{R}^3$ cannot span $\mathbb{R}^3$.

We saw that $\text{span}(\vec{v}, \vec{w})$ may be $\{\vec{0}\}$, a line through the origin, or a plane through the origin, but will never be the whole of $\mathbb{R}^3$.

Problem

Do *three* vectors span $\mathbb{R}^3$? Sometimes? Always? Never?

Let's look at some examples.

Examples 7.

a) The three unit vectors $\vec{i}, \vec{j}$ and $\vec{k}$ do span $\mathbb{R}^3$, because *any* vector

$$\vec{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

in $\mathbb{R}^3$ can be written as a linear combination of $\vec{i}, \vec{j}, \vec{k}$, namely,

$$\vec{b} = b_1 \vec{i} + b_2 \vec{j} + b_3 \vec{k}.$$

Spanning sets for $\mathbb{R}^3$

Problem

So do three vectors *always* span $\mathbb{R}^3$?

b) Consider three vectors which are scalar multiples of each other, for example,

$$\vec{u} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}, \quad \vec{v} = \begin{pmatrix} 2 \\ -4 \\ 6 \end{pmatrix}, \quad \vec{w} = \begin{pmatrix} -5 \\ 10 \\ -15 \end{pmatrix}.$$

The span of these vectors is only a line, not the whole of $\mathbb{R}^3$. Therefore $\{\vec{u}, \vec{v}, \vec{w}\}$ is not a spanning set for $\mathbb{R}^3$.

Spanning sets for $\mathbb{R}^3$

c) **Harder:** Do the vectors

$$\vec{u} = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}, \quad \vec{v} = \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix}, \quad \vec{w} = \begin{pmatrix} -2 \\ -3 \\ 7 \end{pmatrix}$$

span $\mathbb{R}^3$?

Answer. We take an arbitrary vector $\vec{b}$ in $\mathbb{R}^3$ and attempt to write it as a linear combination

$$\vec{b} = \lambda_1 \vec{u} + \lambda_2 \vec{v} + \lambda_3 \vec{w}.$$

Obviously we can find $\lambda_1, \lambda_2, \lambda_3$ in some cases (for example, when $\vec{b} = \dots$?)
But the important question is, can we find $\lambda_1, \lambda_2, \lambda_3$ in *all* cases, no matter what $\vec{b}$ is? Writing the vector equation in the form of three scalar equations, we have

$$\lambda_1 - 2\lambda_2 - 2\lambda_3 = b_1, \quad \lambda_2 - 3\lambda_3 = b_2, \quad -2\lambda_1 + 4\lambda_2 + 7\lambda_3 = b_3.$$

We put these equations into augmented matrix form and row-reduce:

$$(A|\vec{b}) = \left(\begin{array}{ccc|c} 1 & -2 & -2 & b_1 \\ 0 & 1 & -3 & b_2 \\ -2 & 4 & 7 & b_3 \end{array} \right) \longleftrightarrow (U|\vec{y}) = \left(\begin{array}{ccc|c} 1 & -2 & -2 & b_1 \\ 0 & 1 & -3 & b_2 \\ 0 & 0 & 3 & 2b_1 + b_3 \end{array} \right).$$

Spanning sets for $\mathbb{R}^3$, continued

$$(U|\mathbf{y}) = \left(\begin{array}{ccc|c} 1 & -2 & -2 & b_1 \\ 0 & 1 & -3 & b_2 \\ 0 & 0 & 3 & 2b_1 + b_3 \end{array} \right) \quad \bullet \text{ No matter what } b_1, b_2 \text{ and } b_3 \text{ are, the last column } \mathbf{y} \text{ is not a leading column, so the system has at least one solution;}$$

- therefore, the system has a solution not only for some right hand sides, but for **every** right hand side;
- therefore, $\{\vec{\mathbf{u}}, \vec{\mathbf{v}}, \vec{\mathbf{w}}\}$ is a spanning set for $\mathbb{R}^3$.

Comment. It is strongly recommended that in this type of problem you give a three-part reason for your answer: (i) what you see in the echelon form; (ii) what this says about solutions of the system of equations; (iii) how this answers the question asked.

Spanning sets, another example.

Exercise 8. Do the vectors

$$\vec{u} = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}, \quad \vec{v} = \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix}, \quad \vec{w} = \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}$$

span $\mathbb{R}^3$? If not, find condition(s), if any, on b_1, b_2, b_3 for $\vec{b}$ to be in the span of $\vec{u}, \vec{v}, \vec{w}$. Give a detailed geometric interpretation of $\text{span}\{\vec{u}, \vec{v}, \vec{w}\}$.



Solution of example 8.

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> with(LinearAlgebra):
M := <<1,-1,3>|<-1,3,1>|<2,-3,4>|<b1, b2, b3>>;
      M :=  $\begin{bmatrix} 1 & -1 & 2 & b_1 \\ -1 & 3 & -3 & b_2 \\ 3 & 1 & 4 & b_3 \end{bmatrix}$ 
> GaussianElimination(M);
       $\begin{bmatrix} 1 & -1 & 2 & b_1 \\ 0 & 2 & -1 & b_2 + b_1 \\ 0 & 0 & 0 & b_3 - 5b_1 - 2b_2 \end{bmatrix}$ 

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$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & b_1 \\ -1 & 3 & -3 & b_2 \\ 3 & 1 & 4 & b_3 \end{array} \right) \begin{array}{l} \times(-3) \\ \times(1) \end{array} \xrightarrow[\begin{array}{l} R_2 \leftarrow R_2 + R_1 \\ R_3 \leftarrow R_3 - 3R_1 \end{array}]{\begin{array}{l} \text{pink} \\ \text{green} \end{array}} \left(\begin{array}{ccc|c} 1 & -1 & 2 & b_1 \\ 0 & 2 & -1 & b_1 + b_2 \\ 0 & 4 & -2 & -3b_1 + b_3 \end{array} \right) \begin{array}{l} \times(-2) \\ \times 1 \end{array}$$

$$\xrightarrow{R_3 \leftarrow R_3 - 2R_2} \left(\begin{array}{ccc|c} 1 & -1 & 2 & b_1 \\ 0 & 2 & -1 & b_1 + b_2 \\ 0 & 0 & 0 & -5b_1 - 2b_2 + b_3 \end{array} \right)$$

Span of sets of vectors in $\mathbb{R}^m$ and matrices

Preparation for the big result to come.

Exercise 9.

Consider the matrix $A = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}$ and the vector $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$.

Fill in the blanks: $A\vec{x} = x_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + x_2 \begin{pmatrix} 3 \\ 4 \end{pmatrix} + x_3 \begin{pmatrix} 5 \\ 6 \end{pmatrix}$.



In plain English, this result says that $A\vec{x}$ can be interpreted as a linear combination where

- the vectors in the linear combination are the columns of A ,
- the coefficients of the linear combination are the components of $\vec{x}$.

Span of sets of vectors in $\mathbb{R}^m$ and matrices

TIP!

The key here is to keep in mind that $\vec{b} = A\vec{x}$, means that $\vec{b}$ is a linear combination of the columns of A , with scalar coefficients $x_1, x_2, \dots, x_n$, the components of $\vec{x}$.

Fact (Matrices, linear combinations and spans in $\mathbb{R}^m$.)

Let $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ be vectors in $\mathbb{R}^m$, and let A be the $m \times n$ matrix whose columns are $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$. Then

- a specific vector $\vec{b} \in \mathbb{R}^m$ is a linear combination of the vectors $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ if and only if $\vec{b}$ can be written $\vec{b} = A\vec{x}$ for some vector $\vec{x}$ in $\mathbb{R}^n$;
- a specific vector $\vec{b} \in \mathbb{R}^m$ belongs to $\text{span}(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n)$ if and only if the equation $A\vec{x} = \vec{b}$ has a solution in $\mathbb{R}^n$.
- Corollary: $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ spans $\mathbb{R}^m$ if and only if $A\vec{x} = \vec{b}$ has a solution for every $\vec{b}$ in $\mathbb{R}^m$.

Span of sets of vectors in $\mathbb{R}^m$ and matrices

Definition (column space of a matrix)

Let A be an $m \times n$ matrix with entries in a field $\mathbb{F}$. The span of the columns of A is called the **column space** of A and is denoted by $\text{col}(A)$.

In other words, if A has columns $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$, then $\text{col}(A)$ and $\text{span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ are just two names for the same thing.

Note that since $\text{col}(A)$ is the span of certain vectors in $\mathbb{F}^m$, it is a subspace of $\mathbb{F}^m$.

Systems of linear equations (MATH1131 revision)



■ A matrix is in **row echelon form** (REF) if:

- all rows of zeros are at the bottom, and
- each leading entry is further to the right than all leading entries in the rows above it,

where the **leading entry** of a non-zero row is the leftmost non-zero entry in that row. A column containing a leading entry is called a **leading column**.

■ The **nature of solutions** of a system of linear equations $A\mathbf{x} = \mathbf{b}$ (for some $\mathbf{b} \in \mathbb{R}^m$, $\mathbf{x} \in \mathbb{R}^n$, and $A \in M_{m,n}(\mathbb{R})$) can be found once its corresponding augmented matrix $(A|\mathbf{b})$ is reduced to **row echelon form** $(U|\mathbf{y})$.

- If the last column $\mathbf{y}$ is a leading column, then the system has **no solutions**.
- Otherwise, if the last column $\mathbf{y}$ is not a leading column, the system has at least one solution:
 - If all of the columns of U are leading columns, then the system has a **unique solution**.
 - If any of the columns of U are not leading columns, then the system has **infinitely many solutions**. If there are exactly k non-leading columns in U , then the solution set is determined by exactly k **parameters**.